Predicting Accident Severity Using Weather and Road Infrastructure Data

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***Abstract*—Accurately predicting accident severity is critical for improving road safety and reducing fatalities. This study leverages traditional machine learning techniques to analyze accident data from the state of Florida, focusing on the impact of weather conditions, road infrastructure, and temporal factors. The data preprocessing pipeline included handling missing values, addressing class imbalance using SMOTE, and engineering features like *Twilight Phase*. A *Random Forest Classifier* was employed, achieving an accuracy of 88.00% and a mean *F1 Macro Score* of 84.57% across 5-fold cross-validation. Feature importance analysis revealed that factors such as *Distance(mi)*, *Temperature(F)*, and *Traffic Signal* significantly influence accident severity. These findings provide actionable insights for policymakers to mitigate severe accidents and enhance road safety. Future work aims to explore advanced models and additional features to improve prediction performance further.**

***Index Terms*—Accident severity, machine learning, class imbalance, SMOTE, Random Forest, road safety, weather conditions, feature importance.**

I. Introduction

Accident severity prediction is a critical challenge with significant implications for road safety, public policy, and urban planning. Understanding the factors contributing to the severity of road accidents can help authorities develop targeted interventions to mitigate risks, enhance infrastructure, and optimize resource allocation for emergency response.

The primary objective of this project is to develop a predictive model that estimates the severity of an accident based on historical data from the United States [1], focusing on features such as weather conditions, road infrastructure. The problem is approached as a multiclass classification task, where the severity of accidents is categorized into four levels. By leveraging machine learning techniques like Random Forest with class weighting and SMOTE for data balancing, the project addresses the inherent class imbalance and strives to achieve interpretable results.

The dataset comprises information like including weather parameters (e.g., temperature, humidity, visibility, and precipitation), road characteristics (e.g., distance, crossings, and traffic signals), and temporal details (e.g., Day or Night and twilight phase). Preprocessing steps such as missing value imputation, categorical feature encoding, and balancing through SMOTE were critical in preparing the data for effective analysis.

II. Method

A. Data Preprocessing

The analysis was conducted using accident data filtered specifically for the state of Florida. The preprocessing steps included refining the dataset by retaining only critical columns deemed relevant for predicting accident severity. These critical features included *Severity, Distance(mi), Temperature(F), Humidity(%), Visibility(mi), Precipitation(in), Weather_Condition*, and several binary indicators related to road conditions and infrastructure. These binary indicators encompassed features such as *Bump, Crossing, Junction, Stop, Traffic_Calming, Traffic_Signal, Sunrise_Sunset*, and *Twilight_Phase*.

1) Twilight Phase Feature Engineering: To simplify the dataset and enhance interpretability, the columns related to different twilight phases (*Civil_Twilight, Nautical_Twilight, Astronomical_Twilight*) were combined into a single binary feature, *Twilight_Phase*. This new feature indicates whether the accident occurred during any twilight condition, providing a unified perspective on twilight-related impacts on accident severity.

2) Understanding and Reducing Data Distribution: An initial analysis of the class distribution revealed a significant imbalance, with *Severity 2* accounting for 85.83% of all entries, followed by *Severity 3* at 11.87%, *Severity 4* at 1.50%, and *Severity 1* at 0.80%. This distribution indicated that the dataset was dominated by *Severity 2*, leading to the potential underrepresentation of other classes.

To address this imbalance and refine the dataset:

- Rows corresponding to *Severity 4* with missing values were entirely removed due to their limited representation. This ensured the remaining data was complete for this class.
- Missing values for other features, such as *Precipitation(in)*, *Temperature(F)*, and *Visibility(mi)*, were handled conditionally for *Severity 1, 3, and 4*.
  - Correlations among *Temperature(F), Humidity(%), Visibility(mi), Precipitation(in)*, and *Distance(mi)* were analyzed to explore direct relationships with *Severity*. No significant correlations were observed.

  - Conditional imputation was applied based on logical groupings:
    * *Precipitation(in)* values were imputed using median values within each *Weather_Condition* group.
    * *Temperature(F)* values were imputed using medians grouped by *Severity*.
    * *Visibility(mi)* was filled based on medians within *Weather_Condition* groups.
  - *Severity 2* rows were excluded from imputation as the focus was on reducing rows in this class to address the imbalance.
- Following the removal of missing data, the distribution remained highly skewed towards *Severity 2* (83%). The next step involved identifying duplicate rows within the *Severity 2* class to reduce overrepresentation. Duplicate rows were detected using all columns except the target variable, *Severity*.
- After removing duplicates in *Severity 2*, the class distribution improved significantly:
  - *Severity 2*: 75.25%
  - *Severity 3*: 20.73%
  - *Severity 4*: 2.61%
  - *Severity 1*: 1.39%

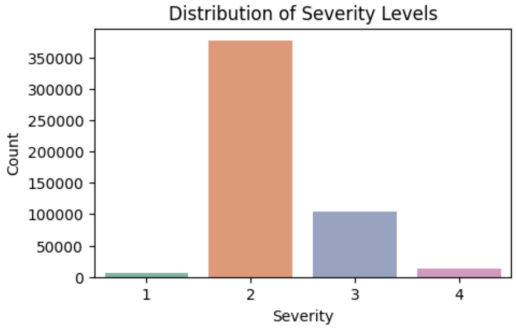


Fig. 1. Class Distribution After Removing Duplicates

- Finally, categorical features (*Weather_Condition, Sunrise_Sunset*) were transformed into numeric representations using one-hot encoding, ensuring compatibility with machine learning algorithms.

B. Addressing Class Imbalance

Despite improvements, the dataset still exhibited class imbalance. To further balance the data, *SMOTE (Synthetic Minority Oversampling Technique)* was applied. This technique was used to generate synthetic samples for the minority classes. The target distribution for SMOTE was set as follows:

- *Severity 1*: 25% of the *Severity 2* count
- *Severity 3*: Maintained at its original count
- *Severity 4*: 20% of the *Severity 2* count
- *Severity 2*: Unchanged

After applying SMOTE, the class distribution was as follows:

- *Severity 1*: 94,280 samples
- *Severity 2*: 377,123 samples
- *Severity 3*: 103,910 samples
- *Severity 4*: 75,424 samples

C. Model Training and Evaluation

The preprocessed dataset was split into training and testing subsets using an 80:20 stratified split to preserve class proportions. A *Random Forest Classifier* was selected for its robustness, interpretability, and ability to handle mixed data types. To address class imbalance during training, class weights were calculated inversely proportional to the frequency of each class in the training set and applied during model training.

1) Evaluation Metrics: The model was evaluated using:

- *F1-Score (Macro and Weighted Averages)*: To balance precision and recall across all classes.
- *Confusion Matrix*: To analyze misclassification patterns.
- *Cross-Validation*: A 5-fold cross-validation was performed to validate the robustness of the model across multiple splits.

These metrics provided insights into the model's performance across the highly imbalanced dataset.

III. Results and Analysis

A. Model Performance

The trained *Random Forest Classifier* achieved an overall **accuracy of 88.00%**. Detailed performance metrics, including precision, recall, and F1-score for each class, are presented in Table I. The results demonstrate a balanced performance across all severity levels, with strong precision and recall for the majority class (*Severity 2*) and significant improvements in minority classes (*Severity 1, 3, and 4*) due to SMOTE and weighted training.

TABLE I
Classification Report for the Random Forest Model

Class	Precision	Recall	F1-Score	Support
Severity 1	91.96%	92.29%	92.12%	18,856
Severity 2	88.98%	94.58%	91.70%	75,425
Severity 3	82.12%	72.57%	77.05%	20,782
Severity 4	84.35%	70.95%	77.07%	15,085
Accuracy	88.00%			
Macro Avg	86.85%	82.60%	84.49%	130,148
Weighted Avg	87.78%	88.00%	87.72%	130,148

The confusion matrix, shown in Table II, highlights the distribution of true and predicted classifications. *Severity 2* demonstrates high recall and precision, while *Severity 3* and *Severity 4* exhibit minor misclassifications, predominantly with adjacent severity levels.

B. Cross-Validation Results

To validate the robustness of the model, a 5-fold cross-validation was performed. The *F1 Macro Score* across the folds was consistently high, as shown in Table III. The average F1 Macro Score of **84.57%** confirms the reliability and generalizability of the model.

TABLE II
CONFUSION MATRIX FOR THE RANDOM FOREST MODEL

True / Pred	Severity 1	Severity 2	Severity 3	Severity 4
Severity 1	17,402	836	518	100
Severity 2	508	71,337	2,294	1,286
Severity 3	911	4,189	15,082	600
Severity 4	103	3,807	472	10,703

TABLE III
CROSS-VALIDATION F1 MACRO SCORES

Fold	F1 Macro Score
Fold 1	84.59%
Fold 2	84.62%
Fold 3	84.48%
Fold 4	84.59%
Fold 5	84.58%
Mean	84.57%

C. Feature Importance

The top 10 features contributing to the model's predictions are listed in Table IV. The most influential feature was *Distance(mi)* with a relative importance of 37.63%, followed by *Temperature(F)* (17.36%) and *Humidity(%)* (15.22%). Features related to weather conditions (*Fair Weather*, *Partly Cloudy*) and road infrastructure (*Traffic Signal*, *Crossing*) also had significant impact.

TABLE IV
TOP 10 FEATURES BY IMPORTANCE

Feature	Importance (%)
Distance(mi)	37.63%
Temperature(F)	17.36%
Humidity(%)	15.22%
Weather_Condition_Fair	5.53%
Traffic_Signal	4.17%
Crossing	2.47%
Visibility(mi)	2.25%
Precipitation(in)	1.82%
Sunrise_Sunset_Night	1.71%
Weather_Condition_Partly Cloudy	1.40%

D. Interpretation

The results demonstrate the model's ability to predict accident severity with high accuracy and balanced performance across all severity levels. The feature importance analysis highlights the critical role of weather conditions and road infrastructure in determining accident severity. *Distance(mi)* emerged as the most significant factor, likely reflecting the impact of high-speed roads on severe accidents. Similarly, environmental factors like *Temperature(F)* and *Humidity(%)* suggest a strong correlation between weather and accident outcomes.

The consistent cross-validation results validate the model's robustness, while the confusion matrix provides insights into areas where misclassification occurs, particularly between adjacent severity levels (*Severity 3* and *Severity 4*). These findings provide actionable insights for policymakers and traffic authorities to mitigate accident severity through targeted interventions.

IV. CONCLUSION

This study explored the application of traditional machine learning techniques to predict accident severity using weather conditions, road infrastructure, and time-related features. The project addressed a real-world problem with significant societal relevance, aiming to enhance road safety and inform policymakers through actionable insights.

The data preprocessing pipeline successfully managed challenges like class imbalance, missing values, and the incorporation of domain-relevant features such as *Twilight Phase*. The use of *SMOTE* and class weighting ensured balanced representation of all severity classes, enabling the model to focus on minority classes like *Severity 1* and *Severity 4*, which are critical for mitigating severe accidents.

The *Random Forest Classifier* demonstrated robust performance, achieving an overall accuracy of **88.00%** and a mean *F1 Macro Score* of **84.57%** across 5-fold cross-validation. These results highlight the model's ability to generalize well across unseen data and provide balanced performance across all severity levels.

Key findings from feature importance analysis underscored the significance of environmental factors such as *Temperature(F)*, *Humidity(%)*, and *Visibility(mi)*, alongside road infrastructure indicators like *Distance(mi)*, *Traffic Signal*, and *Crossing*. These insights emphasize the interplay of weather conditions and road characteristics in influencing accident severity, providing valuable guidance for targeted interventions.

In conclusion, this study demonstrates the potential of traditional machine learning techniques in addressing critical societal challenges like road safety. By leveraging data-driven insights, stakeholders can implement proactive measures to reduce accident severity and improve public safety.

V. CONTRIBUTION

A. Joshna sindhuja Pitta

- Conducted initial data exploration, including filtering the dataset for Florida and identifying critical features for analysis.
- Designed and implemented the missing value imputation strategy, including conditional imputations and the application of *SimpleImputer*.
- Engineered the *Twilight Phase* feature by consolidating twilight-related columns.
- Performed one-hot encoding of categorical variables and prepared the dataset for machine learning.

B. Naga Pujitha Pitta

- Handled class imbalance through the removal of duplicates and the application of *SMOTE*.
- Implemented and optimized the *Random Forest Classifier*, including setting up class weights for handling imbalance during training.

- Conducted cross-validation experiments and evaluated model performance using metrics such as F1-Score and confusion matrix.
- Analyzed feature importance and generated insights from the top predictors of accident severity.

C. Joint Contributions

- Collaborated on designing the experimental framework and interpreting the results.
- Co-authored the report, with equal involvement in documenting the methodology, results, and analysis.

REFERENCES

[1] S. Moosavi, "US Accidents (2016 - 2023)," [Online]. Available: https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents.